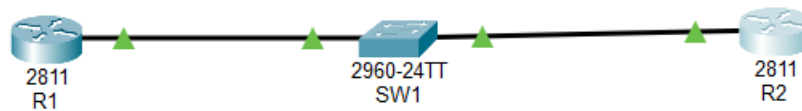


Week 5: Cisco Router and Switch Basics (10 Marks)



28.R1

```
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
```

29.R2

```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R2
```

30. SW1

```
Switch>ena
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW1
```

31. R1

```
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#interface f0/0
R1(config-if)#ip address 10.10.10.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
```

32. R2

```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R2
R2(config)#interface f0/0
R2(config-if)#ip address 10.10.10.2 255.255.255.0
R2(config-if)#no shutdown
R2(config-if)#exit
```

33. SW1

```
Switch>ena
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW1
SW1(config)#interface vlan 1
SW1(config-if)#ip address 10.10.10.10 255.255.255.0
SW1(config-if)#no shutdown
```

34. pinging R2 on SW1

```
SW1#ping 10.10.10.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:
..!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/13/55 ms

SW1#
```

35.

36.

```
R1#show cdp
Global CDP information:
  Sending CDP packets every 60 seconds
  Sending a holdtime value of 180 seconds
  Sending CDPv2 advertisements is enabled
R1#
```

37.

```
show cdp
Global CDP information:
  Sending CDP packets every 60 seconds
  Sending a holdtime value of 180 seconds
  Sending CDPv2 advertisements is enabled
R1#show cdp neighbor
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID        Local Intrfce  Holdtme    Capability  Platform  Port ID
R1#
```

38.

```
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#no cdp run
R1(config)#cdp run
R1(config)#
```

39.

```
R1>ena
R1#show cdp neighbor
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID        Local Intrfce  Holdtme    Capability  Platform  Port ID
R1#
```

40.

```
R2>ena
R2#show ip interface brief
Interface          IP-Address      OK? Method Status
Protocol
FastEthernet0/0    10.10.10.2      YES manual up
FastEthernet0/1    unassigned      YES unset  administratively down down
Vlan1              unassigned      YES unset  administratively down down
R2#
```

41.

```
R2(config)#int F0/0
R2(config-if)#shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to administratively
down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state
to down

R2(config-if)#exit
R2(config)#show ip interface brief
^
% Invalid input detected at '^' marker.

R2(config)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console
show ip interface brief
Interface          IP-Address      OK? Method Status
Protocol
FastEthernet0/0    10.10.10.2      YES manual administratively down down
FastEthernet0/1    unassigned      YES unset  administratively down down
Vlan1              unassigned      YES unset  administratively down down
R2#
```

42.

```
R2(config)#int F0/0
R2(config-if)#no shutdown

R2(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state
to up
show ip interface brief
^
% Invalid input detected at '^' marker.

R2(config-if)#end
R2#
%SYS-5-CONFIG_I: Configured from console by console

R2#show ip interface brief
Interface          IP-Address      OK? Method Status
Protocol
FastEthernet0/0    10.10.10.2      YES manual up
FastEthernet0/1    unassigned      YES unset  administratively down down
Vlan1              unassigned      YES unset  administratively down down
R2#
```

43.

```
SW1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW1(config)#int F0/2
SW1(config-if)#duplex half
SW1(config-if)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#show int F0/2
FastEthernet0/2 is up, line protocol is up (connected)
  Hardware is Lance, address is 0090.2bc0.1402 (bia 0090.2bc0.1402)
  BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Half-duplex, 100Mb/s
  input flow-control is off, output flow-control is off
  ARP type: ARPA, ARP Timeout 00:04:00
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    956 packets input, 193351 bytes, 0 no buffer
    Received 956 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 watchdog, 0 multicast, 0 pause input
    0 input packets with dribble condition detected
    2357 packets output, 263570 bytes, 0 underruns
--More-- |
```

44.

```
>SW1#
SW1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW1(config)#int F0/2
SW1(config-if)#duplex full
```

45.

```
>SW1#
SW1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SW1(config)#int F0/2
SW1(config-if)#duplex full
SW1(config-if)#speed 10
SW1(config-if)#end
SW1#
%SYS-5-CONFIG_I: Configured from console by console

SW1#show int F0/2
FastEthernet0/2 is up, line protocol is up (connected)
  Hardware is Lance, address is 0090.2bc0.1402 (bia 0090.2bc0.1402)
  BW 100000 Kbit, DLY 1000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 10Mb/s
  input flow-control is off, output flow-control is off
  ARP type: ARPA, ARP Timeout 00:04:00
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    956 packets input, 193351 bytes, 0 no buffer
    Received 956 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 watchdog, 0 multicast, 0 pause input
    0 input packets with dribble condition detected
    2357 packets output, 263570 bytes, 0 underruns
--More-- |
```

46.

```
R2>ena
R2#show ip interface
FastEthernet0/0 is up, line protocol is up (connected)
  Internet address is 10.10.10.2/24
  Broadcast address is 255.255.255.255
  Address determined by setup command
  MTU is 1500 bytes
  Helper address is not set
  Directed broadcast forwarding is disabled
  Outgoing access list is not set
  Inbound access list is not set
  Proxy ARP is enabled
  Security level is default
  Split horizon is enabled
  ICMP redirects are always sent
  ICMP unreachable are always sent
  ICMP mask replies are never sent
  IP fast switching is disabled
  IP fast switching on the same interface is disabled
  IP Flow switching is disabled
  IP Fast switching turbo vector
  IP multicast fast switching is disabled
  IP multicast distributed fast switching is disabled
  Router Discovery is disabled
--More--
```

47. the mismatch causes communication chaos because the two devices don't agree on how to share the medium.